

Practical 4

```
In [12]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: x=np.array([90,80,56,76,97])
```

```
In [3]: y=np.array([85,78,90,65,70])
```

```
In [4]: model=np.polyfit(x , y , 1)
```

```
In [5]: model
```

```
Out[5]: array([ -0.30831974, 102.20391517])
```

```
In [6]: predict = np.poly1d(model)
```

```
In [7]: predict(65)
```

```
Out[7]: 82.16313213703097
```

```
In [8]: y_pred=predict(x)
y_pred
```

```
Out[8]: array([74.45513866, 77.53833605, 84.93800979, 78.77161501, 72.29690049])
```

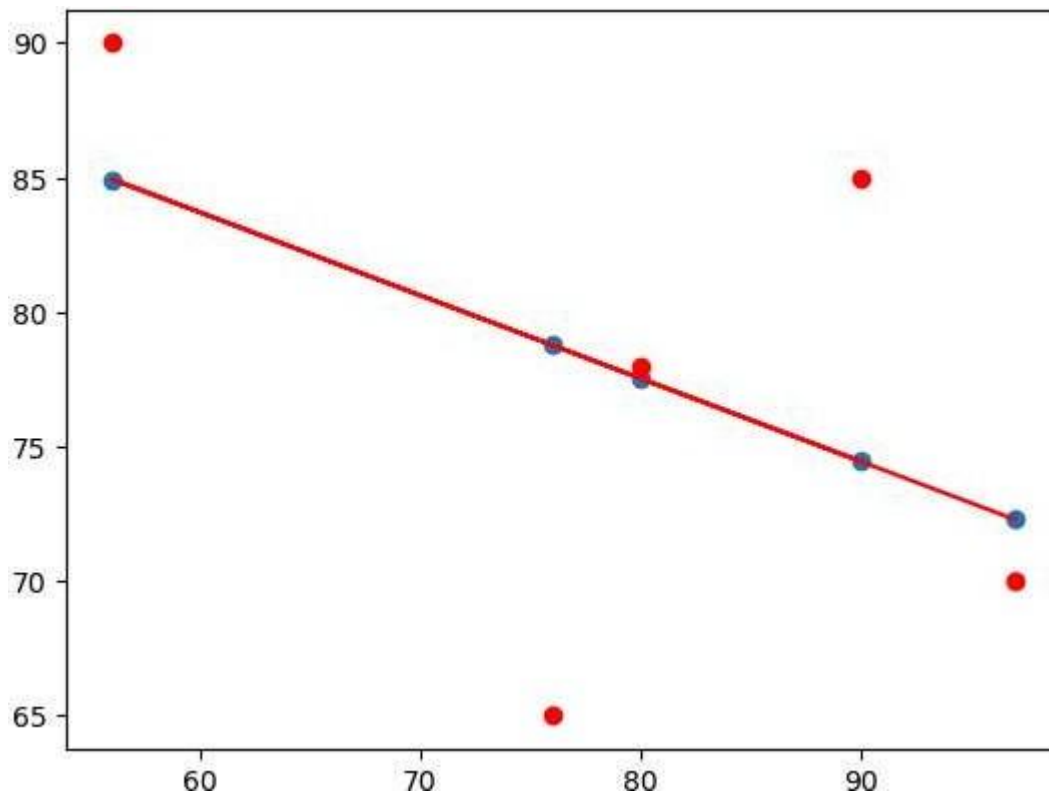
```
In [9]: from sklearn.metrics import r2_score
```

```
In [10]: r2_score(y, y_pred)
```

```
Out[10]: 0.2192753741066471
```

```
In [13]: y_line = model[1]+model[0]*x
plt.plot(x,y_line, c = 'r')
plt.scatter(x,y_pred)
plt.scatter(x,y,c='r')
```

```
Out[13]: <matplotlib.collections.PathCollection at 0x175cf0e9bd0>
```



```
In [23]: import pandas as pd
import numpy as np
df = pd.read_csv(r'C:\Users\System21\Downloads\housing.csv')
```

```
In [25]: df.head()
```

```
Out[25]:
```

	RM	LSTAT	PTRATIO	MEDV
0	6.575	4.98	15.3	504000.0
1	6.421	9.14	17.8	453600.0
2	7.185	4.03	17.8	728700.0
3	6.998	2.94	18.7	701400.0
4	7.147	5.33	18.7	760200.0

```
In [29]:
```

```
In [30]: y = df['RM']
```

```
y
```

```
Out[30]: 0      6.575
         1      6.421
         2      7.185
         3      6.998
         4      7.147
         ...
        484     6.593
        485     6.120
        486     6.976
        487     6.794
        488     6.030
        Name: RM, Length: 489, dtype: float64
```

```
In [31]: df.isnull().sum()
```

```
Out[31]: RM      0
        LSTAT    0
        PTRATIO  0
        MEDV     0
        dtype: int64
```

```
In [32]: x = df.drop(['RM'], axis = 1)
        y = df['RM']
```

```
In [33]: y
```

```
Out[33]: 0      6.575
         1      6.421
         2      7.185
         3      6.998
         4      7.147
         ...
        484     6.593
        485     6.120
        486     6.976
        487     6.794
        488     6.030
        Name: RM, Length: 489, dtype: float64
```

```
In [34]: from sklearn.model_selection import train_test_split
```

```
In [36]: xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size =0.2,random_state =
```

```
In [37]: import sklearn
        from sklearn.linear_model import LinearRegression
        lm = LinearRegression()
        model=lm.fit(xtrain, ytrain)
        ytrain_pred = lm.predict(xtrain)
        ytest_pred = lm.predict(xtest)
```

```
In [39]: df=pd.DataFrame(ytrain_pred,ytrain)
        df=pd.DataFrame(ytest_pred,ytest)
```

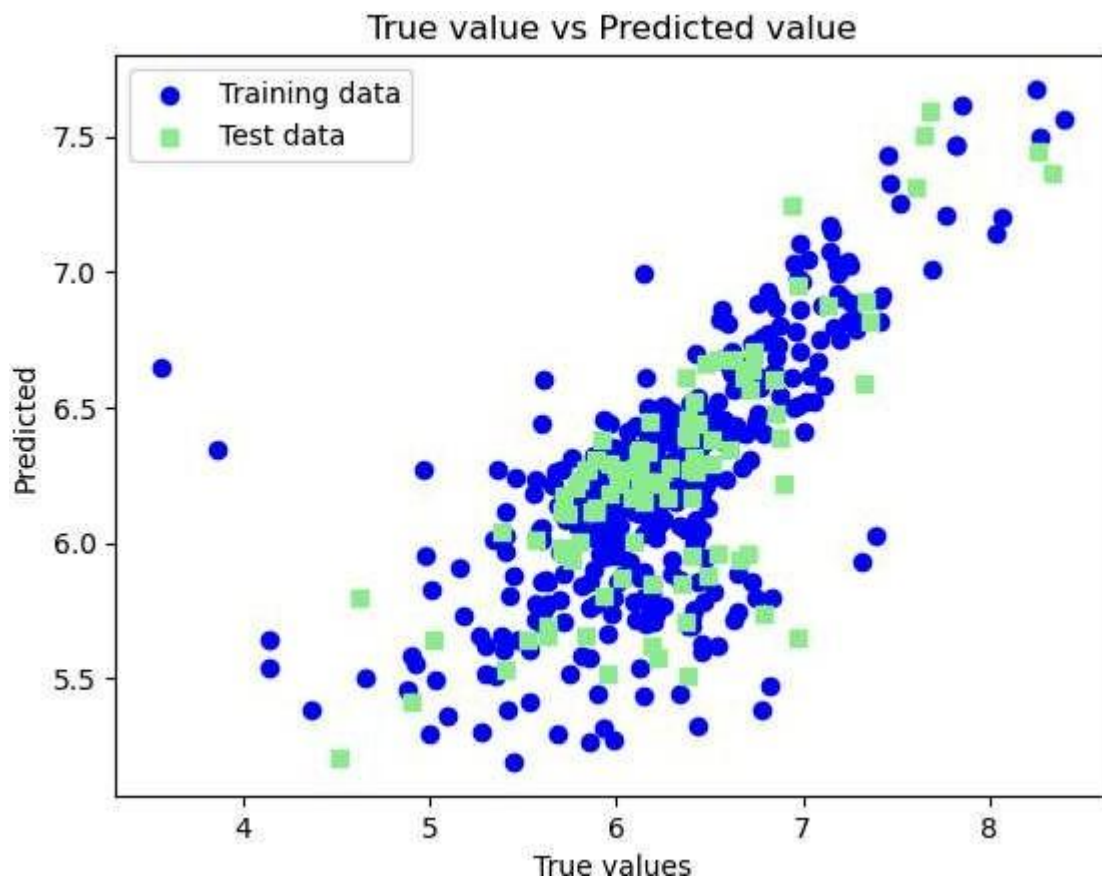
```
In [40]: from sklearn.metrics import mean_squared_error, r2_score
```

```
In [42]: mse = mean_squared_error(ytest, ytest_pred)
print(mse)
mse = mean_squared_error(ytrain_pred,ytrain)
print(mse)
```

0.17705459629995898

0.21183432057206536

```
In [44]: plt.scatter(ytrain ,ytrain_pred,c='blue',marker='o',label='Training data')
plt.scatter(ytest,ytest_pred ,c='lightgreen',marker='s',label='Test data')
plt.xlabel('True values')
plt.ylabel('Predicted')
plt.title("True value vs Predicted value")
plt.legend(loc= 'upper left')
plt.plot()
plt.show()
```



```
In [ ]:
```